Mid-Tenure Research Statement

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Research Vision

Every culture has its acoustic folklore. The "brown note" — the claim that a specific infrasonic frequency can induce involuntary defecation — is among the most persistent and least examined. NASA tested it obliquely during the Apollo programme, subjecting volunteers to sound pressure levels of 140 dB to 154 dB without observing gastrointestinal effects. *South Park* devoted an episode to it (Parker and Stone, "The Brown Noise", S3E17, 1999). Yet no prior study had quantitatively compared airborne and mechanical excitation pathways for abdominal cavity modes within a unified shell-theoretic framework. We found that oversight irresistible.

What began as that single analysis has grown into a ten-paper research programme. The founding question — can airborne sound make the abdomen resonate? — proved to be the wrong question, and wrong questions are often the most productive. The abdomen *does* resonate, at frequencies squarely within the ISO 2631 band of peak human sensitivity to whole-body vibration (4 Hz to 10 Hz). But airborne sound cannot meaningfully excite those modes: at infrasonic frequencies the acoustic wavelength exceeds 85 m, the Helmholtz number $ka \approx 0.011$ implies negligible scattering efficiency, and the resulting coupling penalty suppresses airborne excitation by over four orders of magnitude relative to mechanical vibration. The brown note, as popularly conceived, is a physical impossibility.

This negative result opened the investigation rather than closing it. If whole-cavity resonance cannot be driven acoustically, what about intraluminal gas pockets? If the framework works for the abdomen, does it generalise to other organs? To other species? To a watermelon? Can the same physics that disproves folklore explain the sounds clinicians hear through a stethoscope? And — perhaps most unexpectedly — can the shape and stiffness of an unknown cavity be recovered from its resonant frequencies alone?

A Distinguished Advisory Board convened to evaluate the programme's trajectory crystallised the identity that had been emerging across these questions: **how geometry mediates excitation, resonance, and identifiability in fluid-filled viscoelastic shells**. The Board's framing was characteristically direct: "The brown note is your Rayleigh's blue sky — an entry point, not the legacy." Just as Rayleigh's question about why the sky is blue led to the foundations of scattering theory, the brown note question has led to a general spectral mechanics of soft shell resonance with implications across biomechanics, diagnostics, agriculture, and inverse problems.

The programme's intellectual contribution is fourfold, centred on **how geometry mediates excitation, resonance, and identifiability in fluid-filled viscoelastic shells**. First, a **transferable analytical framework** — oblate spheroidal shell theory with energy-consistent fluid–structure coupling — applicable to any fluid-filled biological or agricultural cavity. Second, **dimensional analysis and scaling laws** that reveal quasi-universal behaviour across mammalian scales. Third, **closed-form resonance inversion** with rigorous identifiability analysis, proving when geometry can and cannot be recovered from spectral data. Fourth, and most unexpectedly, a demonstration that **forward model adequacy does not guarantee inverse model adequacy** — the central insight of the identifiability work, with implications extending well beyond the specific systems studied. The target venues reflect this breadth: the *Journal of Sound and Vibration*, the *Journal of the Acoustical Society of America*, *Postharvest Biology and Technology*, *Inverse Problems in Science and Engineering*, and — for the planned capstone synthesis — the *Proceedings of the Royal Society A*.

The Four Conjectures

The programme converges on four conjectured results that, once proved in the capstone, would establish the mathematical core of a general framework:

1. **Rank-deficiency of scalar-proxy models.** Any forward model mapping geometry to frequency through a scalar proxy necessarily produces a rank-deficient Jacobian for the inverse problem.
2. **Identifiability lifting by asphericity.** Oblate eccentricity introduces mode-dependent curvature sampling that breaks the scalar-proxy degeneracy and restores full-rank identifiability.
3. **Near-spherical conditioning asymptotics.** As eccentricity vanishes, the condition number diverges algebraically, classifying the intrinsic difficulty of inversion for near-spherical cavities.
4. **Forward adequacy ≠ inverse adequacy.** A forward model can predict frequencies to arbitrary accuracy while being provably non-identifiable for parameter recovery.

Accomplishments to Date

The programme has produced ten papers in a single extended research campaign, forming a coherent arc: from whole-cavity analysis, through local inclusions and cross-species scaling, to organ extension, clinical acoustics, sub-bass perception, agricultural diagnostics, inverse-problem identifiability, and a unifying curvature-lifting theorem.

Papers 1–9 form a three-act arc: **Act I** (Papers 1–2) asks whether the brown note exists and identifies why it cannot; **Act II** (Papers 3–6) tests the framework across species, organs, and forcing pathways; **Act III** (Papers 7–9) extracts the theoretical core — inversion, identifiability, and the geometry-mediation thesis.

Paper 1 — Modal Analysis of the Abdominal Cavity

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"
Target: *Journal of Sound and Vibration* (~46 pages). Submission-ready after seven rounds of Reviewer B scrutiny, traversing a REJECT-to-ACCEPT trajectory.

The foundational paper models the human abdomen as a fluid-filled viscoelastic oblate spheroidal shell ($a = 0.18$ m, $c = 0.12$ m, $h = 10$ mm, $E = 0.1$ MPa). The analysis predicts a lowest flexural resonance at $f_2 \approx 4$ Hz, with higher modes at $f_3 \approx 6.3$ Hz and $f_4 \approx 8.9$ Hz — precisely within the ISO 2631 band. These predictions are consistent with experimentally observed whole-body resonance frequencies reported by Kitazaki and Griffin (1998) and the ISO 2631 standard. The breathing mode lies at approximately 2500 Hz, dominated by fluid bulk modulus and irrelevant to infrasound.

The central quantitative result is the coupling ratio: mechanical whole-body vibration drives abdominal wall displacement $\mathbf{3.3 \times 10^4}$ **times** more effectively than airborne sound at equivalent excitation levels. This is an SDOF upper bound with unit modal participation; correcting for the canonical participation factor ($\Gamma_2 \approx 0.48$) gives $\mathcal{R} \approx 1.6 \times 10^4$ — still decisive. At 120 dB SPL, energy-consistent airborne displacement reaches just 0.028 µm — well below the 0.5 µm to 2.0 µm threshold for PIEZO mechanosensitive channel activation. Reaching 1 µm by airborne excitation alone would require approximately 151 dB, beyond any plausible environmental exposure. By contrast, whole-body vibration at occupational exposure levels produces displacements of several millimetres. At the EU action value of $0.5\,\mathrm{m\,s^{-2}}$, predicted displacements approach $\xi/h \sim 0.4$, entering the regime where geometric nonlinearity becomes relevant (Alijani and Amabili, 2014). This strengthens rather than weakens the coupling asymmetry:

the airborne pathway remains deeply linear. The energy-consistent displacement formulation avoids the 6.5× overestimate that arises from naïve pressure-based calculations.

A 10,000-sample Monte Carlo uncertainty quantification identifies elastic modulus as the dominant parameter (Sobol total-order index $S_T = 0.86$), with 90% of configurations placing f_2 within 3.3 Hz to 15.5 Hz. A six-layer multilayer wall model predicts $f_2 = 4.6$ Hz, within 16% of the homogeneous baseline.

Paper 2 — Gas Pockets as Acoustic Transducers

Bowel Gas as an Acoustic Transducer: A Constrained Bubble Model for Infrasound-Induced Mechanotransduction in the Gastrointestinal Tract
Target: *JASA* (~18 pages). Submission-ready.

Where Paper 1 demonstrated that whole-cavity resonance cannot be driven acoustically, Paper 2 identifies the most plausible alternative pathway: intraluminal gas pockets acting as local acoustic transducers. Using a Church (1995) encapsulated-bubble formulation adapted for intestinal wall properties, the model predicts that gas pockets of 5 mL to 100 mL produce tissue-wall displacements **35–100 times** larger than the whole-cavity flexural mode at the same incident SPL. A 100 mL pocket reaches the PIEZO activation threshold at approximately 111 dB; the whole-cavity pathway requires over 180 dB. A fatal equation flaw discovered during internal review — the constrained-bubble formulation did not recover the Minnaert free-bubble limit — was corrected, restoring physical consistency.

Paper 3 — Cross-Species Scaling Laws

Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: Predictions Across Mammalian Scales
Target: *JSV — Short Communication* (~8 pages). Under revision.

Paper 3 applies Buckingham Pi dimensional analysis to the 10-parameter shell model, reducing the flexural eigenfrequency problem to five governing dimensionless groups. Cross-species predictions for rat ($f_2 = 15.7$ Hz), cat (7.2 Hz), pig (4.5 Hz), and human ($f_2 \approx 4$ Hz) reveal a quasi-universal dimensionless frequency $\Pi_0 \approx 0.07$ across a six-fold range in body size. The airborne scattering penalty $\mathcal{R}_{\text{scat}}$ remains in the 10^3–10^4 range regardless of species, validating animal models for mechanical but not acoustic excitation studies.

Paper 4 — Bladder Resonance

Resonant Frequencies of the Human Urinary Bladder: A Fluid-Filled Viscoelastic Shell Model
Target: *Journal of Sound and Vibration* (~27 pages). Minor revision addressed.

Paper 4 extends the framework to the urinary bladder, whose mechanical properties vary with fill state. The model predicts a characteristic U-shaped frequency curve: geometric softening competes with strain-stiffening of the detrusor muscle, producing a minimum $f_2 = 13.5$ Hz at 222 mL. For physiologically relevant fill volumes (150 mL to 500 mL), the $n = 2$ mode spans 13.5 Hz to 17.1 Hz. The mechanical-to-airborne coupling ratio is $\mathcal{R} \approx 6{,}400$ — smaller than the abdominal value owing to the bladder's reduced size, but still decisive. The paper offers the first mechanistic explanation for vibration-induced urinary urgency.

Paper 5 — Borborygmi

What Pitch Is a Growling Stomach? A Unified Multi-Mode Acoustic Model of Borborygmi
Target: *JASA* (~20 pages). Accepted internally.

Paper 5 turns from vibration response to sound *generation.* A five-mechanism model — free Minnaert oscillation, tissue-constrained bubble resonance, Helmholtz resonance through peristaltic constrictions, axial piston modes, and radial breathing of gas columns — predicts bowel sound frequencies from measurable anatomical parameters with no fitted coefficients. The tissue-constrained bubble model predicts 135 Hz to 440 Hz for gas pockets of 1 mL to 50 mL, substantially overlapping the clinically observed healthy band of 200 Hz to 550 Hz. Wall stiffness proves secondary: a 100-fold increase in elastic modulus shifts the constrained frequency by only 9%.

Paper 6 — Sub-Bass Perception at Concerts

Can You Feel the Bass? Quantitative Partition of Airborne and Structure-Borne Pathways for Sub-Bass Perception at Concert Sound Pressure Levels
Target: *JASA* (~18 pages). Accepted internally.

Paper 6 asks whether the "feel it in your gut" experience at concerts can be caused by airborne sub-bass alone. At 40 Hz and 115 dB SPL, airborne displacement reaches only 0.35% of the perception threshold, while structure-borne vibration through floors and seats is approximately 2,600× larger and reaches 9× threshold. Across the full 1 Hz to 100 Hz band, airborne displacement *never* exceeds 1% of the perception threshold. The paper reframes the question from "can bass be felt?" to "which pathway is actually doing the work?"

Paper 7 — Watermelon Ripeness

Can You Hear the Ripeness? Non-Destructive Acoustic Assessment of Fruit Maturity via Equivalent-Sphere Shell Resonance Inversion
Target: *Postharvest Biology and Technology* (~32 pages). Submission-ready.

Paper 7 demonstrates that the framework generalises beyond biology to agriculture. The traditional watermelon "thump test" is reformulated as a physics-based inverse problem: the fruit is modelled as a fluid-filled viscoelastic shell whose tap-tone frequency can be inverted in closed form to recover rind elastic modulus, which tracks ripeness through pectin degradation. The predicted tap-tone drops from 203 Hz (unripe) to 46 Hz (overripe), a 4.4× reduction. The key insight — that f_2^2 is effectively linear in rind modulus ($R^2 > 0.9999$) — makes the inversion simple and well-conditioned. A geometric invariant $\Pi_{\text{ripe}} = 0.062 \pm 0.002$ holds across four cultivars. Paper 7 proved crucial to the programme's intellectual development: the ease of the watermelon inversion threw into sharp relief the difficulty of the organ inversion, motivating Paper 8's identifiability analysis.

Paper 8 — Kac Identifiability

Can You Hear the Shape of an Organ? Practical Identifiability of Viscoelastic Shell Parameters from Resonant Frequencies
Target: *Inverse Problems in Science and Engineering* or *JSV* (~29 pages). Submission-ready after eleven rounds of Reviewer B scrutiny.

Paper 8 reframes the programme's inverse problem as a Kac-style question: given measured resonant frequencies, can one recover the geometry and stiffness of an unknown organ? The answer has two layers. **Layer 1** (chain-rule rank deficiency) proves that if frequencies depend on geometry only through the equivalent radius $R_{\text{eq}} = (a^2c)^{1/3}$, then the Jacobian is rank-deficient and the system is fundamentally non-identifiable. The scaled condition number in the spherical limit is $\kappa = 1.37 \times 10^{10}$. **Layer 2** (mode-dependent curvature sampling) shows that oblate eccentricity introduces mode-dependent corrections that break this degeneracy, reducing the condition number to $\kappa = 69.4$ at the canonical oblate geometry. The gap from 10^{10} to 69 reflects two distinct effects: replacing the scalar-proxy model architecture (responsible for most of the improvement) and introducing oblate eccentricity (which provides a further factor of ~ 4 via mode-dependent curvature sampling). The practical noise floor is $\kappa \approx 269$.

The paper's most far-reaching result is the **forward–inverse adequacy gap**: a model can be perfectly adequate for forward prediction while being catastrophically ill-conditioned for inversion. The equivalent-sphere approximation predicts frequencies to within a few percent of the oblate model, yet it is provably non-identifiable for the inverse problem. This distinction has implications for any field that uses simplified forward models as a basis for parameter estimation.

Paper 9 — Curvature Lifting Theorem

Working title: On the Curvature-Lifting Mechanism in Fluid-Filled Shell Inversion
Target: *Journal of Sound and Vibration — Short Communication.*

Paper 9 is a focused communication proving the curvature-lifting theorem: that oblate eccentricity restores full-rank identifiability by introducing mode-dependent curvature sampling. It extracts the core mathematical result from Paper 8 and proves it in generality, independent of the specific application.

Paper 10 — Capstone Synthesis

Working title: Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells
Target: *Proceedings of the Royal Society A* (scaffolded).

The programme's capstone paper formalises the mathematical results that emerged across Papers 1–9 into four conjectured results (see Section 4). Its backbone is the synthesis of Papers 1, 7, and 8: the abdomen establishes the framework, the watermelon reveals the forward–inverse contrast, and the identifiability analysis provides the mathematical core. Papers 2–6 and 9 serve as supporting evidence for the generality of the conjectures. The target venue reflects the ambition to position this as a contribution to applied mathematical mechanics rather than a domain-specific review.

Methodological Contributions

Analytical Toolchain

The programme's primary methodological contribution is a **modular analytical framework** for vibroacoustic resonance of fluid-filled biological and agricultural cavities, comprising five interlocking components:

1. **Oblate spheroidal shell theory** with Kirchhoff–Love kinematic assumptions within a Rayleigh–Ritz energy framework, following Lamb (1882) and Junger & Feit (1986), with fluid added-mass coupling and viscoelastic loss (Papers 1, 3, 4, 6). The energy-consistent formulation avoids the 6.5× overestimate that arises from pressure-based displacement calculations.
2. **Church (1995) constrained-bubble model**, adapted from ultrasound contrast agents to intestinal gas pockets at infrasonic frequencies (Paper 2).
3. **Buckingham Pi dimensional analysis** yielding exact scaling laws that reduce high-dimensional parameter spaces to governing dimensionless groups (Papers 3, 7).
4. **Energy-consistent coupling comparison**, providing a unified metric $\mathcal{R}$ for quantifying mechanical versus airborne excitation across organs, frequencies, and exposure scenarios (Papers 1, 3, 4, 6).
5. **Closed-form resonance inversion** with analytical identifiability proofs based on mode-dependent curvature sampling and chain-rule rank analysis (Papers 7, 8, 9).

Computational Infrastructure

The programme operates through infrastructure designed for reproducibility and self-consistency: **487 passing tests** covering analytical modules, figure generation, and parameter consistency; **279 merged pull requests** across 549 commits, each reviewed before integration; **automated three-reviewer panels** (domain significance, adversarial rigour, computational reproducibility), with papers undergoing up to eleven review rounds; and **cross-paper consistency auditing** to keep canonical parameters synchronised across all nine manuscripts. Ninety-two research logs document the programme's progress with quantitative detail.

AI-Assisted Research Methodology

The programme constitutes a methodological experiment in its own right. Twenty-four specialised AI agents and 22 reusable skills — spanning paper writing, data analysis, simulation engineering, literature monitoring, experimental protocol design, and adversarial review — operate within a structured semester-based workflow. The human faculty supervisor provides research direction, domain scepticism, and quality judgement; the AI system handles derivation, implementation, drafting, figure generation, and review logistics. This is not a replacement for expertise but a **compression of research overhead**, enabling a ten-paper portfolio to emerge from approximately 30 hours of wall-clock interaction. This is not a claim of autonomy. The system cannot set its own research questions, judge the significance of its results, or decide when a paper is good enough. Those remain irreducibly human functions. But between the question and the judgement lies an enormous volume of skilled labour — derivation, implementation, drafting, checking, revising — and it is there that the compression is real.

Strategic Direction

Advisory Board Assessment

The programme's Distinguished Advisory Board — comprising perspectives from fundamental physics, computational mathematics, science communication, and institutional strategy — convened to evaluate the body of work. Their verdict was **"promising but unfocused"**. The Board identified genuine strengths in depth, breadth, and novelty, while noting that the programme's lasting impact depends on consolidation — extracting the unifying mathematical structure from the individual applications. The Board's central message was that the programme had produced enough individual results to constitute a significant body of work, but had not yet made that extraction.

The Board proposed a one-sentence programme identity: *how geometry mediates excitation, resonance, and identifiability in fluid-filled viscoelastic shells.* They framed the programme's trajectory with an analogy: "The brown note is your Rayleigh's blue sky." The entry point was whimsical; the destination is serious mechanics.

The Four Conjectures

The Board identified four conjectured results latent in the existing papers, each of which we believe can be stated and proved in generality beyond the specific applications:

1. **Rank-deficiency of equivalent-radius models.** Any forward model that maps geometry to frequency through a scalar proxy $g(a, c)$ necessarily produces a rank-deficient Jacobian for the inverse problem. Generalised from Paper 8's equivalent-radius result to arbitrary scalar geometric reductions.
2. **Identifiability lifting by asphericity.** Oblate eccentricity introduces mode-dependent curvature sampling that breaks the rank degeneracy. The geometry-only improvement is a factor of ~ 4; the

remaining improvement reflects replacing the scalar-proxy model architecture. Asphericity is the minimal geometric perturbation that restores full rank — the "curvature lifting theorem."

3. **Asymptotic conditioning near the spherical limit.** As eccentricity $\varepsilon \to 0$, the condition number diverges algebraically, and the rate of divergence classifies the difficulty of the inverse problem for near-spherical cavities.
4. **Forward adequacy ≠ inverse adequacy.** A forward model can predict frequencies to arbitrary accuracy while being provably non-identifiable for parameter recovery. The watermelon (well-conditioned) versus the organ (ill-conditioned) provides the concrete illustration; the theorem is general.

The Capstone Paper

The Board's primary recommendation was a capstone synthesis paper for the *Proceedings of the Royal Society A* — not a review, but a **framework paper** that extracts the four conjectures, proves them in generality, and positions the preceding papers as evidence. The backbone is Papers 1 + 7 + 8: the abdomen establishes the physics, the watermelon reveals the forward–inverse contrast, and the identifiability analysis provides the mathematical machinery. New computation required includes a forward-error-versus-inverse-conditioning figure, generalised rank-deficiency proofs for any scalar $g(a, c)$, curvature-sampling diagnostics per mode, and a cross-application classification table.

The working title — *Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells* — doubles as the programme's one-sentence identity. The key insight, distilled: "The joke was never the point. Geometry was."

Consolidation Priorities

Three operational priorities follow from the Board's assessment:

1. **Stop adding applications unless they test a conjecture.** The programme has sufficient breadth; what it needs is depth. New systems should be pursued only if they advance or test one of the four conjectures.
2. **Seek one phantom collaboration.** A silicone oblate spheroidal phantom instrumented with laser vibrometry would provide the first external test of the analytical predictions, anchoring the programme in empirical reality.
3. **Monograph after capstone, not before.** The temptation to write a monograph prematurely has been noted and resisted. The capstone must come first to establish the intellectual spine around which the full narrative can be organised.

Research Trajectory

Near-Term: Submission Wave

Four papers are submission-ready: Paper 1 (JSV, ~46 pp.), Paper 2 (JASA, ~18 pp.), Paper 7 (Postharvest Biology and Technology, ~32 pp.), and Paper 8 (IPSE or JSV, ~29 pp.). Papers 3 and 4 require minor revisions. Papers 5 and 6 have passed the internal three-reviewer panel and await final formatting. The immediate priority is to begin the external peer review process across multiple venues simultaneously.

Medium-Term: Consolidation and Validation

The capstone synthesis paper (Paper 10) is the programme's central medium-term objective, with an estimated timeline of six to eight weeks to submission quality. In parallel, two experimental

collaborations are being explored: an oblate spheroidal silicone phantom ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$) to test the predicted $f_2 \approx 4\,\mathrm{Hz}$, and clinical bowel sound recordings from healthy subjects to test Paper 5's frequency predictions against measured spectra.

Long-Term: From Programme to School

The Advisory Board's "obituary line" for the programme reads: *He began by asking whether the brown note existed, and ended by founding a general spectral mechanics of fluid-filled soft shells.* Achieving this requires three further steps: the capstone paper (establishing the mathematical core), at least one experimental validation (anchoring the theory in measurement), and a monograph (collecting the full programme into a coherent volume). The long-term applications — non-invasive elastography, clinical bowel sound analysis, agricultural quality assessment, acoustic metamaterial analogies — follow naturally from a validated framework, and need not be pursued until the theoretical foundations are secure.

Doctoral Projects

The programme generates at least three natural doctoral projects: (a) experimental validation of the predicted modal spectrum using silicone oblate spheroidal phantoms instrumented with laser Doppler vibrometry; (b) extension to non-oblate geometries — prolate shells, irregular anatomies, and coupled multi-cavity systems — testing the generality of the curvature-lifting mechanism; and (c) computational inversion using clinical bowel sound recordings or watermelon tap-tone data, bridging the gap between analytical prediction and real-world measurement.

Limitations

The programme is purely analytical: no experimental validation has yet been performed, no finite-element modelling has been employed beyond mesh-convergence verification, and the framework has not been tested against irregular geometries or clinical data. These represent future work rather than present deficiencies — the analytical foundations must be secure before experimental extension is meaningful.

Broader Impact

From Folklore to Framework

The programme demonstrates that folklore can seed rigorous science. The brown note hypothesis led to the first systematic analysis of abdominal vibroacoustic resonance, and thence to a ten-paper portfolio spanning cavity mechanics, clinical acoustics, agricultural diagnostics, and inverse identifiability theory. The lesson is not that folklore is reliable, but that the effort required to *disprove* it rigorously often produces frameworks of independent value.

Generalisable Framework

The analytical toolchain is not specific to any single organ or fruit. Any fluid-filled cavity — biological, agricultural, or industrial — can be modelled within the same framework. The scaling laws provide principled extrapolation between species and systems. The identifiability analysis generalises to any inverse problem in which a forward model's geometric dependence passes through a low-dimensional proxy — a structure common across biomedical imaging, non-destructive testing, and geophysical inversion.

AI-Assisted Research as Method

The programme is, to our knowledge, among the first to employ a structured multi-agent AI system as the primary research executor under human supervision. The infrastructure — 24 agents, 22 skills, 92 research logs, and an open-source repository — offers a replicable template for other research groups. The central insight is that AI's comparative advantage lies not in replacing scientific judgement but in **compressing the overhead** that separates a research idea from a publishable manuscript.

Reflections on Process

The read-to-write ratio across the system exceeds 1:1 — the agents spend as much time inspecting and reviewing their own output as they do generating it. This is not an artefact; it was designed in.

The σ_{abs} Story

During Paper 1's seventh Reviewer B round, a factor-of-four error was discovered in the Breit–Wigner absorption cross-section formula. This was not a typographical issue; it was a genuine physics bug that had propagated through the energy-consistent displacement calculation, affecting the coupling ratio and the airborne displacement estimates throughout the programme. Correcting it shifted ξ_{energy} from 0.014 µm to 0.028 µm and $\mathcal{R}$ from 6.6×10^4 to 3.3×10^4. The qualitative conclusions were unchanged — airborne coupling remains negligible by any measure — but the quantitative precision improved materially.

The episode illustrates two things. First, the adversarial review process works: an AI reviewer, given a clear mandate to find fatal flaws, identified a real error in the underlying physics that had survived multiple earlier rounds. Second, the correction *strengthened* the programme's credibility rather than undermining it. By catching and correcting a factor-of-four error internally, the team demonstrated the same self-correcting rigour that external peer review is designed to provide.

Selected Publications

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1. J. Mace and B. R. Mace, "Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the 'Brown Note'," *Journal of Sound and Vibration*, in preparation, 2026. (~46 pp.)
2. J. Mace and B. R. Mace, "Bowel Gas as an Acoustic Transducer: A Constrained Bubble Model for Infrasound-Induced Mechanotransduction in the Gastrointestinal Tract," *The Journal of the Acoustical Society of America*, in preparation, 2026. (~18 pp.)
3. J. Mace and B. R. Mace, "Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: Predictions Across Mammalian Scales," *Journal of Sound and Vibration — Short Communication*, in preparation, 2026. (~8 pp.)
4. J. Mace and B. R. Mace, "Resonant Frequencies of the Human Urinary Bladder: A Fluid-Filled Viscoelastic Shell Model," *Journal of Sound and Vibration*, in preparation, 2026. (~27 pp.)
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10. J. Mace and B. R. Mace, "Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells," *Proceedings of the Royal Society A*, in preparation, 2026. (scaffolded)